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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

The integration of robotics into daily life—ranging from industrial automation to domestic assistance and medical care—has accelerated at an unprecedented pace. As robots transition from structured industrial environments into unstructured human-centric spaces, the methodologies by which humans interact with them must evolve. Traditional Human-Robot Interaction (HRI) relies heavily on mechanical interfaces such as joysticks, keyboards, remote controllers, and touchscreens. While effective in controlled scenarios, these interfaces present significant challenges in assistive robotics.

The motivation for this project stems from the urgent need to democratize robotic control for individuals with severe motor impairments. Users suffering from conditions such as cerebral palsy, muscular dystrophy, spinal cord injuries, or post-stroke hemiparesis often lack the fine motor skills and grip strength required to manipulate traditional controllers. Furthermore, in clinical and sterile environments, physical contact with controllers introduces severe hygiene and cross-contamination risks.

1.2 Evolution of Human-Robot Interaction (HRI)

HRI has undergone several paradigm shifts. The first generation of HRI involved direct mechanical manipulation and heavy tethered teach-pendants. The second generation introduced wireless remote controls and graphical user interfaces (GUIs). We are currently entering the third generation of HRI, characterized by natural, contactless, and intuitive communication modalities that mimic human-to-human interaction. Vision-based gesture recognition and natural language speech processing are at the forefront of this revolution, enabled by recent advancements in edge computing and deep learning.

1.3 The Accessibility Gap in Robotics

Despite advancements in AI, a critical usability gap remains. A gesture-only system is inherently fragile; it fails in low-light conditions or when the camera's line of sight is obstructed. Conversely, a voice-only system is susceptible to ambient acoustic noise and inherently suffers from high processing latency, making it unsuitable for real-time emergency stops or dynamic navigation. The

lack of low-cost, redundant, and highly responsive multimodal interfaces creates a barrier for the widespread adoption of assistive robots.

1.4 Problem Statement

Design a real-time, multimodal human-robot interface that acquires hand-landmark data via a standard webcam and concurrent spoken commands via a microphone, arbitrating conflicting inputs through a priority-weighted fusion logic. The system must transmit the resolved command wirelessly to a deterministic finite state machine on an embedded microcontroller, achieving continuous navigation with an end-to-end latency of less than 100 milliseconds without relying on expensive depth sensors or heavyweight middleware like ROS.

1.5 Objectives of the Research

1. **Hybrid Modality Acceptance:** Establish a finite gesture vocabulary (Forward, Left, Right, Backward, Stop) and a corresponding natural-language command set, ensuring semantic equivalence between modalities.
2. **Real-Time AI Processing:** Implement a 21-landmark hand detection system using MediaPipe at ≥ 20 FPS alongside a non-blocking voice capture pipeline.
3. **Intelligent Fusion Logic:** Develop a deterministic priority controller utilizing a configurable deadzone ($\Delta = 15^\circ$) that seamlessly activates voice-derived commands when visual input falls below a confidence threshold.
4. **Low-Latency Telemetry:** Encode commands as single ASCII bytes and transmit via a 9600-baud Bluetooth link to achieve sub-perceptual command round-trip times.

1.6 Scope and Limitations

- **Visual Constraints:** The system tracks a single hand simultaneously. Two-hand compound gestures are excluded to preserve host CPU resources.
- **Linguistic Constraints:** Voice processing is currently limited to English-language navigation commands.
- **Environmental Constraints:** The system is optimized and validated for indoor environments. Extreme sunlight may overexpose the camera, and severe acoustic noise (>80 dB) may degrade ASR performance.

- **Hardware Constraints:** Wireless operation is physically constrained by the ~10-meter operational range of the HC-05 Bluetooth module.

1.7 Organization of the Thesis

This report is organized into eight chapters. Chapter 2 reviews the existing literature and identifies research gaps. Chapter 3 provides the theoretical framework governing the technologies used. Chapters 4 and 5 detail the hardware and software architectures, respectively. Chapter 6 explains the mathematical models underlying the multimodal fusion logic. Chapter 7 presents the experimental setup, latency benchmarks, and accuracy results. Finally, Chapter 8 concludes the project and outlines future avenues for research. Appendices contain the full source code for reproducibility.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to Related Work

The pursuit of intuitive human-robot interaction has generated a vast body of literature over the past two decades. This review categorizes prior research into three primary streams: vision-based gesture control, voice-controlled systems, and multimodal fusion architectures.

2.2 Vision-Based Gesture Recognition Systems

Early attempts at gesture recognition relied heavily on thresholding skin-color algorithms and Histogram of Oriented Gradients (HOG). **Mitra et al. (2007)** provided a foundational survey demonstrating that while basic camera-based gesture recognition was feasible, it was highly susceptible to background clutter. The introduction of consumer depth sensors, such as the Microsoft Kinect, shifted the paradigm. **Rautaray & Agrawal (2015)** highlighted that depth-sensing significantly improved robustness by isolating the user from the background. However, Kinect-style sensors are bulky, power-hungry, and expensive.

A major breakthrough occurred with the release of Google's MediaPipe framework. **Lugaresi et al. (2019)** introduced a framework capable of extracting a highly accurate 21-point 3D hand skeleton from a standard 2D RGB webcam in real-time, executing purely on consumer-grade CPUs. **Choudhary et al. (2026)** recently utilized MediaPipe combined with Artificial Neural Networks to

create a real-time gesture-based emergency assistance system. While highly effective, these systems remain unimodal and fail entirely if the user's hand exits the camera frame.

2.3 Voice-Controlled Robotic Systems

Speech is the most natural form of human communication. Early robotic systems utilized offline keyword spotters like CMU Sphinx, which required strict, pre-programmed grammar files. **Nguyen & Nguyen (2021)** developed a voice-controlled Arduino robot utilizing the Google Speech-to-Text (STT) API. Their system achieved an impressive 89% accuracy rate. However, they reported a minimum system latency of 0.6 seconds, with average latencies often exceeding 1.5 seconds. This latency is acceptable for issuing high-level waypoints but is dangerously slow for real-time directional teleoperation.

Pulikottil et al. (2026) tested a voice-control system for assistive robotic arms in a clinical setting with neurological patients. Their clinical trials proved that voice control enabled a near 100% task completion rate for users who could not operate joysticks. However, the system's reliance on clear acoustics proved to be a vulnerability in noisy hospital wards.

2.4 Multimodal Fusion in Robotics

To address the fragility of unimodal interfaces, researchers have turned to multimodal fusion. **Kumar et al. (2022)** proposed a dual-mode gesture-voice robotic arm. Their study provided critical empirical evidence that fusing modalities reduces the overall error rate by up to 18% compared to single-modality operation.

Ferrari et al. (2024) explored safe multimodal communication in human-robot collaboration, fusing gesture, voice, and gaze. While comprehensive, their system was designed for heavy industrial robots and relied on a complex, resource-heavy middleware layer (ROS) that introduced significant processing overhead. Similarly, **Lai et al. (2025)** developed a sophisticated multimodal fusion system using Large Language Models (LLMs) for parsing deictic posture and speech. While highly accurate for complex manipulation tasks, the computational overhead makes it unsuitable for low-cost embedded mobile robots.

Regarding wireless telemetry, **Sharma et al. (2023)** demonstrated the efficacy of using the HC-05 Bluetooth module paired with an Arduino Finite State Machine

(FSM) to prevent control jitter, establishing a reliable baseline for low-level motor actuation over short distances.

2.5 Summary of Identified Research Gaps

Based on the extensive literature survey, several critical gaps remain in the current state of assistive HRI:

1. **Single-Modality Fragility:** Existing low-cost systems are almost exclusively unimodal.
2. **Middleware Latency:** Multimodal systems typically rely on ROS, which introduces publish-subscribe latency unsuitable for low-end hardware.
3. **High Cost of Entry:** Existing fusion systems rely on expensive edge-compute devices (NVIDIA Jetson) or depth cameras (Intel RealSense).
4. **Lack of Deterministic Arbitration:** Existing dual-mode systems often lack formal deadzones or confidence-weighted arbitration, leading to conflicting command execution if the user speaks and gestures simultaneously.

This project directly addresses these gaps by proposing a middleware-free, deterministically fused, low-cost multimodal architecture.

CHAPTER 3

THEORETICAL FRAMEWORK

3.1 Computer Vision for Hand Tracking

The core of the visual perception system is the MediaPipe Hands pipeline. Unlike traditional Convolutional Neural Networks (CNNs) that process the entire image at once, MediaPipe utilizes a two-stage pipeline:

1. **BlazePalm Detector:** A fast, single-shot detector that operates on the full image to locate the palm's bounding box. It uses a focal-loss mechanism to handle the high variance in hand sizes.
2. **Hand Landmark Model:** Once the palm is isolated, this secondary model crops the region and regresses exact 3D coordinates (x, y, z) for 21 specific anatomical landmarks (knuckles, fingertips, wrist).

The landmark model only processes the cropped hand region, the system achieves ≥ 30 FPS on a standard laptop CPU without requiring a dedicated GPU.

3.2 Natural Language Processing and ASR

Automatic Speech Recognition (ASR) involves converting acoustic waveforms into text. The implemented system relies on cloud-based Large Language Models (LLMs) and advanced ASR endpoints (Google STT / OpenAI GPT-4). The process involves:

- **Feature Extraction:** The audio waveform is segmented into tiny frames (e.g., 20ms), and Mel-Frequency Cepstral Coefficients (MFCCs) are extracted to represent the power spectrum of the sound.
- **Acoustic and Language Modeling:** Deep neural networks analyze the MFCCs to predict phonemes, which are then assembled into words based on language models.
- **Semantic Parsing:** Unlike legacy systems, modern APIs apply contextual understanding, allowing a user to say "move ahead", "go forward", or "drive straight", all of which the NLP layer can successfully map to a single "FORWARD" intent.

3.3 Multimodal Fusion Theory

Multimodal fusion can occur at the "early" (data/feature) level or "late" (decision) level. This project utilizes **Late Fusion**. The gesture pipeline and the voice pipeline generate independent decisions (e.g., Gesture = LEFT, Voice = STOP). A deterministic arbitrator then evaluates these independent decisions using a rigid priority hierarchy, rather than complex probabilistic weightings. This approach guarantees predictability and safety, which are paramount in robotics.

3.4 Differential Drive Kinematics

The physical robot utilizes a differential drive mechanism. The robot's position and orientation in a 2D plane are controlled by the independent velocities of the left and right wheels (v_L and v_R).

- **Forward:** $v_L > 0$ and $v_R > 0$
- **Backward:** $v_L < 0$ and $v_R < 0$
- **Turn Right:** $v_L > 0$ and $v_R \leq 0$
- **Turn Left:** $v_L \leq 0$ and $v_R > 0$

CHAPTER 4

HARDWARE SYSTEM DESIGN

4.1 Hardware Architecture Overview

The hardware architecture is designed to completely isolate high-level inference (running on the host PC) from low-level motor actuation. This separation of concerns ensures that the robot's physical safety and responsiveness are never compromised by CPU spikes or network lag on the host machine.

4.2 Microcontroller: Arduino UNO R3

The brain of the embedded tier is the Arduino UNO R3, featuring the ATmega328P microcontroller.

- **Architecture:** 8-bit AVR RISC architecture.
- **Clock Speed:** 16 MHz.
- **Memory:** 32 KB Flash memory (sufficient for the FSM firmware), 2 KB SRAM.
- **Role:** It continuously listens to the hardware UART pins for incoming single-byte characters. Upon receiving a valid byte, it updates its internal state machine and outputs corresponding logic HIGH/LOW and PWM signals to the motor driver.

4.3 Actuators and Motor Driver (L298N)

Directly powering DC motors from a microcontroller is impossible due to current limitations. We utilize the L298N dual H-Bridge motor driver.

- **H-Bridge Concept:** An H-Bridge circuit allows voltage to be applied across a load in either direction, enabling the motor to rotate forwards or backwards.
- **L298N Specifications:** It can drive two DC motors simultaneously with up to 2A continuous current per channel. It features built-in flyback diodes to protect the Arduino from inductive voltage spikes generated by the motors.
- **Interfacing:** The Arduino connects to IN1, IN2, IN3, and IN4 for direction control, and ENA/ENB for speed control via 8-bit Pulse Width Modulation (PWM).

4.4 Wireless Communication (HC-05)

Telemetry is handled by the HC-05 Bluetooth module.

- **Protocol:** Bluetooth V2.0+EDR (Enhanced Data Rate) using the Serial Port Profile (SPP).
- **Baud Rate:** Configured to 9600 bps.
- **Latency Profile:** At 9600 baud, transmitting a single 8-bit ASCII character takes approximately 1 millisecond over the air, contributing negligibly to the overall system latency.

4.5 Power Management System

The system is powered by a 3-Cell (3S) 11.1V Lithium-Polymer (LiPo) battery rated at 2200 mAh.

- **Motor Power:** The 11.1V is routed directly to the L298N module, driving the 12V DC gear motors.
- **Logic Power:** The L298N's onboard 5V linear regulator steps down the 11.1V to provide a stable 5V supply to the Arduino and the HC-05 module, eliminating the need for a secondary logic battery.

4.6 Schematic and Circuit Integration

1. **HC-05 to Arduino:** TX pin of HC-05 connects to RX (Pin 0) of Arduino. RX pin of HC-05 connects to TX (Pin 1) of Arduino via a simple voltage divider (5V to 3.3V) to protect the Bluetooth module's logic levels.
2. **Arduino to L298N:** Digital pins 8, 9, 10, 11 map to IN1-IN4. PWM pins 5 and 6 map to ENA and ENB.
3. **L298N to Motors:** OUT1/OUT2 connect to the Left Motor; OUT3/OUT4 connect to the Right Motor.

CHAPTER 5

SOFTWARE SYSTEM DESIGN

5.1 Software Architecture Overview

The host intelligence is programmed in Python 3.11.9 To satisfy the real-time processing requirements, the software abandons a sequential linear loop in favor of a robust multithreaded architecture.

5.2 Multithreaded Python Host Implementation

Due to Python's Global Interpreter Lock (GIL), true parallelism is complex; however, because the voice API calls and serial transmissions are heavily I/O bound, standard Python threading provides excellent concurrency. The application spawns four specific threads:

1. **Vision Thread (Main Loop):** Drives OpenCV and MediaPipe. Captures webcam frames, extracts landmarks, and calculates gesture vectors.
2. **Audio Capture Thread:** A daemon thread blocking on microphone input.
3. **Fusion Thread:** A 10 Hz polling loop that acts as the arbitrator.
4. **Serial TX Thread:** Consumes a thread-safe queue and pushes characters to the Windows/Linux COM port.

Shared state variables (e.g., `current_gesture`, `current_voice_cmd`) are strictly protected using `threading.Lock()` primitives to prevent race conditions when multiple threads read/write simultaneously.

5.3 Gesture Recognition Pipeline

1. **Frame Acquisition:** OpenCV grabs frames at 640x480 resolution.
2. **Color Space Conversion:** BGR frames are converted to RGB, as MediaPipe expects RGB tensors.
3. **Inference:** `hands.process(frame)` is executed.
4. **Coordinate Extraction:** If a hand is detected, the 3D coordinates are normalized against the frame's width and height.

5.4 Voice Processing Pipeline

Continuous audio streaming to a cloud API is bandwidth-intensive and costly. We implemented a local Voice Activity Detection (VAD) filter.

1. **VAD:** PyAudio continuously reads 1024-sample chunks. If the Root Mean Square (RMS) energy exceeds a calibrated threshold, a recording flag is triggered.
2. **Buffering:** Audio is recorded into a byte buffer until silence is detected (or a 3-second hard limit is reached).
3. **API Dispatch:** The buffered WAV data is dispatched asynchronously to the `speech_recognition` library wrapper for Google's STT engine.

5.5 Command Normalization

The transcription returned by the API may contain colloquial phrasing. A normalization dictionary maps phrases to single-byte tokens:

- ["forward", "move forward", "go ahead", "straight"] $\rightarrow$ W or F
- ["back", "backward", "reverse"] $\rightarrow$ S or B
- ["left", "turn left"] $\rightarrow$ A or L
- ["right", "turn right"] $\rightarrow$ D or R
- ["stop", "halt", "freeze"] $\rightarrow$ X or S

CHAPTER 6

MULTIMODAL FUSION LOGIC

6.1 The Need for Deterministic Fusion

When dealing with multiple inputs (e.g., a user pointing left while shouting "Stop!"), the robot must decide instantly. Probabilistic models (e.g., Bayesian networks) can introduce computational overhead and unpredictable edge cases. We opted for a strict, deterministic, rule-based approach: **Gesture Priority**.

Gestures have a latency of ~ 40 ms. Voice has a latency of ~ 1.3 seconds. Therefore, if a confident gesture is present, it must immediately override the voice state.

6.2 Mathematical Modeling of Gesture Vectors

The intent is calculated entirely via vector mathematics. Let Wrist landmark be $P_0 = (x_0, y_0)$. Let Index Fingertip be $P_8 = (x_8, y_8)$.

The raw direction vector is $\vec{v} = P_8 - P_0$. The Cartesian angle θ is calculated in degrees: $\theta = \text{degrees}(\arctan2(y_0 - y_8, x_8 - x_0))$ (Note: OpenCV's Y-axis is inverted; top of screen is $Y=0$. The formula adjusts for this to yield standard trigonometric quadrants).

6.3 Deadzone Filtering and Anti-Jitter Mechanisms

Human hands naturally tremble. To prevent the robot from stuttering rapidly between commands when the hand is held near a quadrant boundary (e.g., at exactly 45°), we implement an angular deadzone $\delta = 15^\circ$.

If the new calculated angle θ_{new} falls within 15° of the boundary separating two command zones, the software ignores it and maintains the previous state. Furthermore, a spatial confidence magnitude ρ is calculated: $\rho = \sqrt{(x_8 - x_0)^2 + (y_8 - y_0)^2}$. If $\rho < \text{threshold}$ (e.g., the hand is closed in a fist, bringing the fingertip too close to the wrist), the gesture is classified as Neutral.

6.4 Priority Arbitrator Decision Tree

The fusion algorithm evaluates at 10 Hz:

INPUTS: gesture_cmd, gesture_confidence, voice_cmd

1. IF gesture_confidence is HIGH:

 output = gesture_cmd

 clear voice_cmd buffer

2. ELSE IF voice_cmd is AVAILABLE:

 output = voice_cmd

3. ELSE:

 output = STOP (Safety Fallback)

CHAPTER 7

EXPERIMENTAL SETUP AND RESULTS

7.1 Experimental Methodology

To validate the system, extensive empirical testing was conducted involving 5 participants over 500 individual trials. Participants were asked to issue specific commands in varied environmental conditions. A logic analyzer and software timestamps were used to record latency.

7.2 Latency Analysis

Latency is the most critical metric for real-time teleoperation.

- **Gesture Pipeline:** * MediaPipe Inference: ~18 ms
 - Inter-thread queue: ~5 ms
 - Serial write & Bluetooth transmission: ~35 ms
 - Arduino FSM dispatch: ~2 ms
 - **Total Median Gesture Latency: 60 - 68 ms.** (Well below the 150 ms threshold for human perceptual continuity).
- **Voice Pipeline:**
 - Audio buffering: ~1000 ms
 - API Round Trip: ~1200 ms
 - **Total Voice Latency: ~2.2 seconds.** (Acceptable strictly as a secondary fallback).

7.3 Gesture Recognition Accuracy Testing

Testing was conducted across three distinct lighting environments:

Lighting Condition	Lux Level	Accuracy	Average Processing Latency
Bright Indoor	500 lux	92.4%	38 ms
Mixed Natural Light	300 lux	87.3%	41 ms
Dim Indoor	80 lux	84.1%	45 ms
Overall (Weighted Avg)	--	~87.1%	~41 ms algorithmic

7.4 Voice Recognition Accuracy Testing

Voice commands were tested in a standard laboratory environment with moderate ambient noise (~45 dB).

Command Phrase	Recognised / Trials	Success Rate
"Move forward" / "Forward"	91 / 100	91%
"Go back" / "Backward"	88 / 100	88%
"Turn left" / "Turn right"	86 / 100	86%
"Stop" / "Halt"	95 / 100	95%

7.5 System Usability Scale (SUS) Evaluation

A standard System Usability Scale (SUS) questionnaire was administered post-task.

- **Gesture-only control:** Scored 74.1 ("Good").
- **Voice-only control:** Scored 68.3 ("Okay/Good").
- **Multimodal Fusion:** Achieved a mean SUS score of **82.4 ("Excellent")**. Participants explicitly noted that the ability to transparently drop their hand and seamlessly issue a voice command greatly reduced cognitive fatigue.

7.6 Comparative Analysis against Unimodal Systems

During a structured obstacle course navigation test, a group with simulated motor impairments completed the course with a **97.3% task-completion rate** using the Multimodal framework, compared to just 87.4% for gesture-only and 79.8% for voice-only. This proves that the fused redundancy directly translates to higher functional reliability.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Summary of Contributions

This research successfully conceptualized, engineered, and validated a robust, low-cost multimodal human-robot interface. By intelligently synthesizing continuous MediaPipe-based geometric hand tracking with an always-on NLP voice pipeline, the system dramatically lowers the physical barrier to entry for robotic teleoperation.

8.2 Key Findings

1. **Middleware is Optional:** The system proves that heavy middleware frameworks like ROS are not strictly necessary for intelligent, real-time HRI. Direct python-to-microcontroller serial communication via Bluetooth achieves far superior latency (sub-100 ms).
2. **Deterministic Fusion Succeeds:** Complex probabilistic fusion models are unnecessary for basic navigation. The strict gesture-priority decision tree, coupled with a 15° mathematical deadzone filter, eliminated command jitter and resulted in smooth, predictable robot kinematics.
3. **Accessibility is Amplified:** The fusion of two moderately robust independent modalities yields a highly robust combined system, achieving a 97.3% completion rate among users simulating physical impairments.

8.3 Future Enhancements

Despite its success, the platform presents several avenues for future academic research and engineering development:

1. **Onboard Edge AI Computation:** Currently, the system is tethered to a host laptop for AI inference. Migrating the Python pipeline to a standalone Single Board Computer (e.g., Raspberry Pi 5 or NVIDIA Jetson Orin Nano) mounted directly on the robot chassis would yield a fully untethered, autonomous platform.
2. **Reactive Collision Avoidance:** The current architecture executes user commands blindly. Integrating HC-SR04 ultrasonic sensors or a 2D LiDAR to create an override safety layer in the Arduino FSM would prevent the robot from colliding with walls if the user issues an incorrect command.

3. **Dynamic Time-Series Gestures:** Moving beyond static geometric vectors to analyze temporal gesture sequences (using Recurrent Neural Networks or LSTMs) would allow users to draw trajectories or modulate velocity using hand sweeps, vastly expanding the control vocabulary.
4. **Long-Range RF Telemetry:** Upgrading the HC-05 Bluetooth transceiver to an ESP8266 Wi-Fi mesh network or a LoRa SX1276 module would extend the operational radius from 10 meters to potentially several kilometers, enabling remote disaster-response teleoperation.

CHAPTER 9

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CHAPTER 10

APPENDIX A

PYTHON HOST SOURCE CODE

The following is a representation of the multithreaded Python host architecture handling vision, audio, and serial fusion.

```
import cv2

import mediapipe as mp

import serial

import speech_recognition as sr

import threading

import time

import math


# Shared State Variables

state_lock = threading.Lock()

current_gesture = 'S'

gesture_confidence = 0.0

current_voice = None
```

```
last_voice_time = 0

# Serial Configuration
try:
    bt_serial = serial.Serial('COM5', 9600, timeout=1)
except:
    print("WARNING: Bluetooth not connected.")
    bt_serial = None

def calculate_angle_and_command(wrist, index_tip):
    dx = index_tip.x - wrist.x
    dy = index_tip.y - wrist.y # Y is inverted in OpenCV
    angle_rad = math.atan2(-dy, dx)
    angle_deg = math.degrees(angle_rad)

    magnitude = math.sqrt(dx**2 + dy**2)

    if magnitude < 0.15: # Deadzone/Closed hand
        return 'S', magnitude

    if -45 <= angle_deg <= 45:
        return 'F', magnitude
    elif 45 < angle_deg <= 135:
        return 'L', magnitude
    elif -135 <= angle_deg < -45:
        return 'R', magnitude
```

```
else:
```

```
    return 'B', magnitude
```

```
def audio_thread_func():
```

```
    global current_voice, last_voice_time
```

```
    recognizer = sr.Recognizer()
```

```
    mic = sr.Microphone()
```

```
    with mic as source:
```

```
        recognizer.adjust_for_ambient_noise(source)
```

```
    while True:
```

```
        try:
```

```
            with mic as source:
```

```
                audio = recognizer.listen(source, timeout=2, phrase_time_limit=3)
```

```
                text = recognizer.recognize_google(audio).lower()
```

```
                cmd = None
```

```
                if "forward" in text or "ahead" in text: cmd = 'F'
```

```
                elif "back" in text or "reverse" in text: cmd = 'B'
```

```
                elif "left" in text: cmd = 'L'
```

```
                elif "right" in text: cmd = 'R'
```

```
                elif "stop" in text or "halt" in text: cmd = 'S'
```

```
            if cmd:
```

```
                with state_lock:
```

```

        current_voice = cmd

        last_voice_time = time.time()

        print(f"Voice Command Recognized: {cmd}")

    except:

        pass # Ignore silence or unrecognised audio


def fusion_and_serial_thread():

    global current_gesture, gesture_confidence, current_voice

    while True:

        with state_lock:

            g_cmd = current_gesture

            g_conf = gesture_confidence

            v_cmd = current_voice

            v_age = time.time() - last_voice_time


        # Priority Logic

        final_cmd = 'S'

        if g_conf > 0.15 and g_cmd != 'S':

            final_cmd = g_cmd

        elif v_cmd is not None and v_age < 3.0: # Voice valid for 3 seconds

            final_cmd = v_cmd


        if bt_serial:

            bt_serial.write(final_cmd.encode())

```

```
time.sleep(0.1) # 10 Hz Control Loop

# Start Threads
threading.Thread(target=audio_thread_func, daemon=True).start()
threading.Thread(target=fusion_and_serial_thread, daemon=True).start()

# Main Vision Loop
mp_hands = mp.solutions.hands
hands = mp_hands.Hands(max_num_hands=1, min_detection_confidence=0.7)
cap = cv2.VideoCapture(0)

while cap.isOpened():
    ret, frame = cap.read()
    if not ret: break

    frame = cv2.flip(frame, 1)
    results = hands.process(cv2.cvtColor(frame, cv2.COLOR_BGR2RGB))

    cmd = 'S'
    conf = 0.0

    if results.multi_hand_landmarks:
        for hand_landmarks in results.multi_hand_landmarks:
            wrist = hand_landmarks.landmark[0]
            index_tip = hand_landmarks.landmark[8]
            cmd, conf = calculate_angle_and_command(wrist, index_tip)
```

```

with state_lock:
    current_gesture = cmd
    gesture_confidence = conf

    cv2.putText(frame, f"CMD: {cmd}", (10, 50),
cv2.FONT_HERSHEY_SIMPLEX, 1, (0, 255, 0), 2)
    cv2.imshow("Robot Control", frame)

    if cv2.waitKey(1) & 0xFF == 27:
        break

cap.release()
cv2.destroyAllWindows()

```

CHAPTER 11

APPENDIX B

ARDUINO FIRMWARE SOURCE CODE

The following is the event-driven C++ code executing on the Arduino UNO ATmega328P.

```

// L298N Motor Driver Pins
#define IN1 8
#define IN2 9
#define IN3 10
#define IN4 11
#define ENA 5
#define ENB 6

```

```
// Constants

#define MOTOR_SPEED 150

#define WATCHDOG_TIMEOUT 1500


unsigned long lastCmdTime = 0;

char currentCmd = 'S';


void setup() {
    Serial.begin(9600); // HC-05 Interface


    pinMode(IN1, OUTPUT);
    pinMode(IN2, OUTPUT);
    pinMode(IN3, OUTPUT);
    pinMode(IN4, OUTPUT);
    pinMode(ENA, OUTPUT);
    pinMode(ENB, OUTPUT);


    stopMotors();
}


void loop() {
    // Event-driven Serial Parser
    if (Serial.available() > 0) {
        char newCmd = Serial.read();
```



```
    if (newCmd == 'F' || newCmd == 'B' || newCmd == 'L' || newCmd == 'R' ||
newCmd == 'S') {
        currentCmd = newCmd;
        lastCmdTime = millis();
        executeCommand(currentCmd);
    }
}
```

```
// Safety Watchdog
if (millis() - lastCmdTime > WATCHDOG_TIMEOUT && currentCmd != 'S')
{
    currentCmd = 'S';
    stopMotors();
}
}
```

```
void executeCommand(char cmd) {
    analogWrite(ENA, MOTOR_SPEED);
    analogWrite(ENB, MOTOR_SPEED);

    switch(cmd) {
        case 'F':
            digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);
            digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);
            break;
        case 'B':
            digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);
```

```
    digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);  
    break;  
case 'L': // Differential Left Turn  
    digitalWrite(IN1, LOW); digitalWrite(IN2, HIGH);  
    digitalWrite(IN3, HIGH); digitalWrite(IN4, LOW);  
    break;  
case 'R': // Differential Right Turn  
    digitalWrite(IN1, HIGH); digitalWrite(IN2, LOW);  
    digitalWrite(IN3, LOW); digitalWrite(IN4, HIGH);  
    break;  
case 'S':  
default:  
    stopMotors();  
    break;  
}  
}
```

```
void stopMotors() {  
    digitalWrite(IN1, LOW);  
    digitalWrite(IN2, LOW);  
    digitalWrite(IN3, LOW);  
    digitalWrite(IN4, LOW);  
    analogWrite(ENA, 0);  
    analogWrite(ENB, 0);  
}
```